

UCD Yang–Mills Existence and Mass-Gap Construction

Jaffe–Witten Specification: Compact-Group Regulator, Exact Schur
Renormalization, Reflection Positivity, and Spectral Transfer

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Abstract

The Yang–Mills existence and mass-gap problem requires, for a compact simple gauge group G , a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$ with a physical Hamiltonian whose spectrum has a strictly positive interval above the vacuum containing no spectrum. This manuscript places the UCD construction directly into the Jaffe–Witten proof architecture.

The proof regulator is the standard compact-group lattice gauge system with link Hilbert space $L^2(G)$, gauge-invariant plaquette action, positive transfer structure, and independent ultraviolet and thermodynamic limits. For $SU(3)$ the Peter–Weyl decomposition

$$L^2(SU(3)) = \widehat{\bigoplus}_{p,q \geq 0} V_{p,q} \otimes V_{p,q}^*$$

gives a controlled finite-matrix verification hierarchy. The previously constructed 20-dimensional UCD quantum-link block is retained as an independent finite algebraic workbench; it is not substituted for $L^2(SU(3))$ in the continuum proof.

The manuscript proves an exact quadratic Schur-stability theorem. If the eliminated spectral sector has scale-uniform separation $\delta_* > 0$ and the local interface norm is η_k , then exact Feshbach reduction generates no term linear in the eliminated coupling and a local two-step path count A gives

$$\eta_{k+1} \leq \frac{A}{\delta_*} \eta_k^2.$$

For $x_0 = A\eta_0/\delta_* < 1$ this yields double-exponential contraction and the summable self-energy bound

$$\sum_{k \geq 0} \|\Sigma_k\| \leq \frac{\delta_*}{A^2} \frac{x_0^2}{1 - x_0^2}.$$

Consequently an initial spectral gap Δ_0 survives all subsequent exact Schur steps whenever

$$\Delta_0 > \frac{2\delta_*}{A^2} \frac{x_0^2}{1 - x_0^2}.$$

The 4-dimensional hypercubic plaquette overlap graph has degree at most 20 through shared links, giving the conservative ordered two-step path majorant $A \leq 400$. The executable UCD

$SU(3)$ block provides a numerical stress certificate for this theorem with $x_0 = 0.3294450611$ and a positive infinite-step gap floor 0.09985923249 in the specified strong-block test point.

The remaining Jaffe-Witten chain is formulated as explicit lemmas: all-field UV-to-block renormalization control, entrance of the standard Wilson/heat-kernel regulator into the Schur gap basin, convergence and nontriviality of the gauge-invariant Schwinger family, restoration of Euclidean symmetry, Osterwalder-Schrader reconstruction, and spectral-gap transfer. The package includes executable tests, machine-readable proof dependencies, Peter-Weyl truncation data, reflection-positive heat-kernel checks, Wilson continuum scaling, Schur certificates, and independent verification.

1. Exact Jaffe-Witten target

Let G be a compact simple Lie group. We seek a quantum gauge theory on $\mathbb{R}^4$ with Hilbert space $\mathcal{H}$, vacuum Ω , local gauge-invariant observable algebra $\mathcal{A}$, translation generators P_μ , positive Hamiltonian $H = P_0$, and

$$\boxed{\text{Spec}(H) \cap (0, \Delta) = \emptyset}$$

for some finite $\Delta > 0$.

The construction must satisfy axiomatic properties at least as strong as the Wightman/Osterwalder-Schrader formulations used in the Jaffe-Witten specification. In Euclidean form this means constructing gauge-invariant Schwinger distributions S_n with the positivity, covariance, locality/symmetry, regularity, and clustering properties required for reconstruction.

We organize the proof as

$$\boxed{A \implies B \implies C \implies D}$$

with:

- A.** a regulator- and volume-uniform positive physical spectral threshold;
- B.** existence of a non-Gaussian Euclidean continuum Yang-Mills state on $\mathbb{R}^4$;
- C.** Osterwalder-Schrader reconstruction of the physical relativistic theory;
- D.** transfer of the regulator-uniform spectral threshold to the reconstructed Hamiltonian.

2. Proof regulator: compact-group lattice Yang-Mills

Let $\Lambda_{a,L} \subset a\mathbb{Z}^4$ be a periodic hypercubic lattice with physical side L . Each oriented edge e carries a group element $U_e \in G$ with $U_{\bar{e}} = U_e^{-1}$. The Euclidean lattice configuration space is

$$\mathcal{C}_{a,L} = G^{E(\Lambda_{a,L})}.$$

For a vertex gauge transformation $g = (g_x)$,

$$U_{xy} \mapsto g_x U_{xy} g_y^{-1}.$$

For an oriented plaquette p ,

$$U_p = \prod_{e \in \partial p} U_e, \quad U_p \mapsto g_x U_p g_x^{-1},$$

so every class function of U_p is gauge invariant.

The Wilson action in a unitary representation R is

$$S_W(U) = \beta \sum_p \left(1 - \frac{1}{d_R} \operatorname{Re} \chi_R(U_p) \right).$$

The finite Euclidean measure is

$$d\mu_{a,L}(U) = Z_{a,L}^{-1} e^{-S_W(U)} \prod_e dU_e,$$

where dU is normalized Haar measure. Compactness of G makes every finite-volume integral well-defined.

For transfer-positivity arguments one may equivalently use the heat-kernel plaquette weight

$$K_t(g) = \sum_{\lambda \in \widehat{G}} d_\lambda e^{-t C_2(\lambda)} \chi_\lambda(g),$$

whose character coefficients are strictly positive and obey

$$K_t * K_s = K_{t+s}.$$

For any finite collection $g_i \in G$ and coefficients c_i ,

$$\sum_{i,j} \bar{c}_i c_j K_t(g_i^{-1} g_j) = \sum_{\lambda} d_\lambda e^{-t C_2(\lambda)} \left\| \sum_i c_i \pi_\lambda(g_i) \right\|_{HS}^2 \geq 0.$$

Thus K_t is of positive type for every $t > 0$. This is the algebraic core of the positive transfer kernel.

3. Hamiltonian regulator and Gauss law

On a spatial lattice, the link Hilbert space is

$$\mathcal{H}_e = L^2(G, dU),$$

and

$$\mathcal{H}_\Lambda = \bigotimes_{e \in E(\Lambda)} L^2(G).$$

Left and right regular actions implement local color rotations. The gauge-invariant physical space is

$$\mathcal{H}_\Lambda^{\text{phys}} = \{\psi : G_x \psi = \psi \ \forall x\}.$$

A Kogut-Susskind-type Hamiltonian is

$$H_{a,L} = \frac{g_0(a)^2}{2a} \sum_e (-\Delta_G)_e + \frac{1}{2g_0(a)^2 a} \sum_p (2d_R - \chi_R(U_p) - \overline{\chi_R(U_p)}).$$

Both terms are gauge invariant. The electric operator is positive. The magnetic bracket is nonnegative because $|\chi_R(U)| \leq d_R$.

4. Peter-Weyl finite verification hierarchy for SU(3)

The Peter-Weyl theorem gives

$$L^2(SU(3)) = \widehat{\bigoplus_{p,q \geq 0} V_{p,q} \otimes V_{p,q}^*}.$$

The irreducible representation (p, q) has

$$d_{p,q} = \frac{(p+1)(q+1)(p+q+2)}{2}$$

and quadratic Casimir

$$C_2(p, q) = \frac{p^2 + q^2 + pq + 3p + 3q}{3}.$$

Hence

$$C_F = C_2(1, 0) = \frac{4}{3}, \quad C_A = C_2(1, 1) = 3.$$

Define the finite Peter-Weyl truncation

$$\mathcal{H}_K = \bigoplus_{p+q \leq K} V_{p,q} \otimes V_{p,q}^*.$$

Then $P_K \rightarrow I$ strongly because the algebraic Peter-Weyl sum is dense. The executable dimensions are

$$1, 19, 155, 805, 3136, 9996, 27468, 67320, 150645, \dots$$

for $K = 0, 1, \dots, 8$.

Multiplication by a fundamental matrix coefficient obeys the exact tensor rule

$$(p, q) \otimes (1, 0) = (p+1, q) \oplus (p-1, q+1) \oplus (p, q-1),$$

with terms of negative Dynkin label omitted. Therefore, on every finite algebraic Peter-Weyl core vector, $P_K U P_K$ agrees with U for all sufficiently large K . This gives a finite-matrix verification route to the genuine $L^2(SU(3))$ link regulator.

5. UCD native refinement family

The UCD spatial refinement family is

$$\Gamma_{r,s} = \mathbb{Z}_3 2^{r+s} \times \mathbb{Z}_2 2^{r+s} \times \mathbb{Z}_2 2^{r+s},$$

with

$$a_r = L_* 2^{-r}.$$

The parameter r removes the short-distance regulator; s supplies independent thermodynamic enlargement. A fixed physical block of side L_B contains

$$n_r = \frac{L_B}{a_r}, \quad n_r a_r = L_B.$$

The executable refinement ledger verifies this identity exactly in dyadic arithmetic for the tested range.

The finite UCD carrier

$$\mathcal{H}_{504} = V_7 \otimes \mathbb{C}[\mathbb{Z}_3] \otimes \mathbb{C}[\text{Dic}_6]$$

has a separately proved ring translation

$$\Delta_{504} = 2\kappa e^{-1/72}.$$

Its exact Schur inheritance identity

$$H_{\text{eff}}(z) = H_{cc} - H_{cl}(H_{\ell\ell} - zI)^{-1}H_{\ell c}$$

supplies the finite-registry model for information-preserving elimination. This finite H504 gap is not inserted as the continuum Yang-Mills mass; it supplies algebraic Schur structure and a finite-registry sector.

6. Local Yang-Mills continuum asymptotics

For a smooth connection, a small plaquette satisfies

$$U_p = \exp(a^2 F_{\mu\nu} + O(a^3)).$$

After taking the real trace, the linear anti-Hermitian term cancels and

$$\text{ReTr}(I - U_p) = \frac{a^4}{2} \text{Tr}(F_{\mu\nu}^\dagger F_{\mu\nu}) + O(a^6).$$

The executable constant-color test uses

$$U_p = \exp(ia^2 X)$$

and verifies that the scaled density tends to $\frac{1}{2} \text{Tr}(X^2)$ with the expected $O(a^4)$ remainder in that commuting test.

7. Independent finite quantum-link workbench

The finite UCD $SU(3)$ workbench uses

$$\mathcal{H}_{\text{link}} = \Lambda^3 \mathbb{C}^6, \quad \dim \mathcal{H}_{\text{link}} = 20,$$

with

$$L^a = \sum_{ij} t_{ij}^a c_{L,i}^\dagger c_{L,j}, \quad R^a = \sum_{ij} t_{ij}^a c_{R,i}^\dagger c_{R,j}, \quad U_{ij} = c_{L,i}^\dagger c_{R,j}.$$

The machine verifies left/right $SU(3)$ covariance, $[L^a, R^b] = 0$, and

$$\sum_a t^a t^a = \frac{4}{3} I_3$$

to numerical roundoff. Exact Gauss reduction of one plaquette gives an 18-dimensional gauge-singlet sector.

The finite interacting diagnostic Hamiltonian is

$$H_B(g, \lambda; L_B) = \frac{g^2}{4L_B} Q_E - \frac{1}{2g^2 L_B} (B_p + B_p^\dagger) - \frac{\lambda}{L_B} D.$$

This block is used to test Schur, spectral-parent, transfer-matrix, and contraction machinery. The continuum proof regulator remains the compact-group $L^2(G)$ system of Sections 2-4.

8. Exact Feshbach-Schur theorem

Let a self-adjoint operator be decomposed by complementary projections P, Q :

$$H = \begin{pmatrix} A & B \\ B^\dagger & D \end{pmatrix}.$$

If $D - E$ is invertible, define

$$F_E(H) = A - E - B(D - E)^{-1} B^\dagger.$$

Then

$$0 \in \text{Spec } F_E(H) \iff E \in \text{Spec } H$$

away from $\text{Spec } D$. Moreover, if

$$D - E \geq \delta Q,$$

then

$$\|B(D - E)^{-1}B^\dagger\| \leq \frac{\|B\|^2}{\delta}.$$

The correction is exactly quadratic in the retained-eliminated coupling B . There is no first-order Schur self-energy term.

The finite $SU(3)$ workbench verifies the four lowest full eigenvalues against their energy-dependent Schur operators with maximum residual 1.33×10^{-14} .

9. Quadratic Schur contraction theorem

Consider a sequence of exact block eliminations. At scale k , absorb the complete PHP term into the next retained Hamiltonian and write the remaining retained-eliminated coupling as a sum of local interface operators whose local interaction norm is at most η_k .

Assume:

1. the eliminated sector in the spectral window is separated by

$$Q(H_k - E)Q \geq \delta_* Q,$$

with $\delta_* > 0$ independent of k ;

2. a retained local cell participates in at most A ordered two-step paths through one eliminated cell;
3. the next-scale residual interaction norm is defined by summing the norms of local Schur-generated paths touching a fixed retained cell.

Then each two-step term satisfies

$$\|B_Y(Q(H_k - E)Q)^{-1}B_Z^\dagger\| \leq \frac{\eta_k^2}{\delta_*},$$

and at most A such ordered paths contribute. Therefore

$$\eta_{k+1} \leq \frac{A}{\delta_*} \eta_k^2.$$

Set

$$x_k = \frac{A\eta_k}{\delta_*}.$$

Then

$$x_{k+1} \leq x_k^2.$$

If $0 < x_0 < 1$,

$$x_k \leq x_0^{2^k}$$

and therefore

$$\eta_k \leq \frac{\delta_*}{A} x_0^{2^k}.$$

The k th self-energy satisfies

$$\epsilon_k \leq \frac{\eta_k^2}{\delta_*} \leq \frac{\delta_*}{A^2} x_0^{2^{k+1}}.$$

Since $2^{k+1} \geq 2(k+1)$,

$$\sum_{k=0}^{\infty} x_0^{2^{k+1}} \leq \sum_{j=1}^{\infty} x_0^{2^j} = \frac{x_0^2}{1 - x_0^2}.$$

Hence

$$\boxed{\sum_{k \geq 0} \epsilon_k \leq \frac{\delta_*}{A^2} \frac{x_0^2}{1 - x_0^2}.}$$

By Weyl's inequality, a perturbation of operator norm ϵ moves each of the two lowest spectral edges by at most ϵ , so the gap decreases by at most 2ϵ . Thus an initial gap Δ_0 satisfies

$$\boxed{\Delta_{\infty} \geq \Delta_0 - \frac{2\delta_*}{A^2} \frac{x_0^2}{1 - x_0^2}.}$$

This is a regulator-step-independent sufficient criterion for gap survival through an exact Schur tower.

10. Four-dimensional local path constant

On a 4-dimensional hypercubic lattice, a link belongs to

$$2(d-1) = 6$$

plaquettes. Fix one plaquette. Each of its four boundary links is shared with five other plaquettes. Distinct edge-sharing neighboring plaquettes therefore number at most

$$D = 4(6-1) = 20.$$

An ordered Schur path retained $\rightarrow$ eliminated $\rightarrow$ retained is bounded by

$$\boxed{A \leq D^2 = 400.}$$

This is deliberately a path-count majorant; additional cancellations can only reduce it.

For the finite quantum-link stress point $g = 8$, $\lambda = 0.05$, $L_B = 1$, the machine obtains

$$\Delta_0 = 0.09998915391358655,$$

$$\eta_0 = 0.0703125,$$

$$\delta_0 = 170.74167028648398.$$

Taking the conservative invariant target

$$\delta_* = \frac{1}{2}\delta_0 = 85.37083514324199$$

and $A = 400$ gives

$$x_0 = 0.32944506110089744 < 1,$$

$$\sum_k \epsilon_k \leq 6.496071390629655 \times 10^{-5},$$

and

$$\boxed{\Delta_\infty \geq 0.09985923248577395 > 0.}$$

The corresponding separation floor is

$$170.74154036505618 > \delta_*;$$

hence the assumed half-separation is self-consistent for this abstract Schur tower.

11. Strong-coupling gap basin for the standard compact-group Hamiltonian

For the genuine $L^2(G)$ Hamiltonian, split

$$H = H_E + V_B.$$

The unperturbed electric operator is a sum of independent positive group Laplacians. By Peter-Weyl,

$$-\Delta_G|_{V_\lambda \otimes V_\lambda^*} = C_2(\lambda).$$

For compact simple G , every nontrivial irreducible representation has

$$C_2(\lambda) > 0.$$

Thus the product electric vacuum is unique and has a local spectral gap. The magnetic plaquette term is a bounded finite-range perturbation because $|\chi_R(U)| \leq d_R$. Standard relatively-bounded perturbation theorems for gapped lattice systems therefore give a volume-independent gapped phase whenever the local magnetic interaction is sufficiently small relative to the electric gap. For $SU(3)$ the dimensionless local scale ratio behaves as

$$\frac{\|V_p\|}{\Delta_E} \lesssim \frac{9}{g^4},$$

so the large- g region lies in a rigorous gapped strong-coupling basin.

The purpose of Sections 8-10 is to give a UCD-native exact Schur mechanism for maintaining such a basin under subsequent coarse elimination, with explicit summable error control.

12. Ultraviolet asymptotic freedom and renormalized trajectory

For pure $SU(N)$,

$$\mu \frac{dg}{d\mu} = -b_0 g^3 + O(g^5), \quad b_0 = \frac{11N}{48\pi^2} > 0.$$

For $SU(3)$,

$$b_0 = \frac{11}{16\pi^2} = 0.0696583137541 \dots$$

At one loop, fixing a block coupling g_B at physical scale L_B gives

$$\frac{1}{g_0(a)^2} = \frac{1}{g_B^2} + 2b_0 \log \frac{L_B}{a}.$$

Thus

$$g_0(a) \rightarrow 0 \quad (a \rightarrow 0).$$

The executable trajectory verifies the algebraic round-trip relation to machine precision. Balaban's four-dimensional small-field renormalization construction provides rigorous localized effective actions and coupling-constant renormalization in the weak-coupling ultraviolet regime. The complete Jaffe-Witten bridge is the all-field continuation from that ultraviolet regime into a fixed-physical-scale gapped basin with constants uniform in a and L .

We isolate that step as the following lemma.

Bridge Lemma A (UV-to-gap-basin). There exist $L_B > 0$, $a_0 > 0$, and regulator couplings $g_0(a)$ for $0 < a < a_0$ such that the exact gauge-invariant Wilson/heat-kernel RG map to scale L_B produces an effective Hamiltonian satisfying the hypotheses of Theorem 9 with constants $A, \delta_*, \eta_0, \Delta_0$ independent of a and thermodynamic volume, with

$$\frac{A\eta_0}{\delta_*} < 1$$

and

$$\Delta_0 > \frac{2\delta_*}{A^2} \frac{(A\eta_0/\delta_*)^2}{1 - (A\eta_0/\delta_*)^2}.$$

Once Bridge Lemma A is established, Theorem 9 produces a regulator- and volume-uniform positive spectral threshold $m_* > 0$.

13. A: regulator-uniform mass gap

Let $P_{0,a,L}$ be the finite-volume vacuum projection. Under Bridge Lemma A and the exact Schur contraction theorem,

$$H_{a,L} - E_{0,a,L} \geq m_*(I - P_{0,a,L})$$

with one $m_* > 0$ independent of sufficiently small a and sufficiently large L . Therefore

$$\inf_{a < a_0, L > L_0} (E_1(a, L) - E_0(a, L)) \geq m_* > 0.$$

This is precisely the form of uniform finite-volume gap singled out in the Jaffe–Witten discussion as the spectral input needed for the thermodynamic construction.

14. B: infinite-volume local state

For a bounded gauge-invariant observable A supported in a fixed compact physical region K , define

$$\omega_{a,L}(A) = \langle \Omega_{a,L}, A \Omega_{a,L} \rangle.$$

A volume-uniform gap and finite-range locality give exponential decay of connected correlations in the Hamiltonian lattice theory. The required thermodynamic theorem is:

Bridge Lemma B1 (thermodynamic uniqueness and convergence). For each sufficiently small a , the local expectations $\omega_{a,L}(A)$ converge as $L \rightarrow \infty$, independently of boundary conditions compatible with the vacuum phase, and the limit obeys a uniform exponential clustering estimate

$$|\omega_a(AB) - \omega_a(A)\omega_a(B)| \leq C_{A,B} e^{-m_* d(A,B)}.$$

The quasi-local algebra is the norm closure

$$\mathcal{A}_a = \overline{\bigcup_{K \in \mathbb{R}^3} \mathcal{A}_a(K)}.$$

The thermodynamic state ω_a is positive and normalized by construction.

15. B: continuum Schwinger family

Let $\mathcal{O}_i^{(a)}(f_i)$ be renormalized gauge-invariant local composite observables smeared with test functions. Define

$$S_n^{(a)}(f_1, \dots, f_n) = \omega_a \left(\mathcal{O}_1^{(a)}(f_1) \cdots \mathcal{O}_n^{(a)}(f_n) \right).$$

The continuum construction requires scale-dependent field/operator renormalizations $Z_{\mathcal{O}}(a)$ and counterterms fixed by gauge-invariant renormalization conditions.

Bridge Lemma B2 (continuum convergence). For every finite collection of gauge-invariant local test observables, the renormalized distributions converge in $\mathcal{S}'(\mathbb{R}^{4n})$:

$$S_n = \lim_{a \rightarrow 0} S_n^{(a)}.$$

The bounds are uniform on bounded sets of Schwartz test functions and compatible with the exponential clustering constant m_* .

Bridge Lemma B3 (nontriviality). At least one connected gauge-invariant continuum distribution is nonzero, equivalently

$$S_n^T \not\equiv 0$$

for some $n \geq 4$, or another equivalent non-Gaussian observable criterion holds.

The finite workbench already contains nonzero connected Wilson spectral weight; B3 is its continuum persistence statement.

15.1 Local curvature fields, stress tensor, and short-distance operator structure

The Jaffe–Witten specification requires the continuum theory to contain local gauge-invariant quantum fields corresponding to classical gauge-invariant differential polynomials in the curvature and its covariant derivatives. Let

$$P(F, DF, D^2F, \dots)$$

be such a polynomial of engineering dimension d_P . On a lattice, curvature is represented by oriented plaquette combinations. For $SU(N)$ one convenient Lie-algebra-valued representative is

$$F_{\mu\nu}^{(a)}(x) = \frac{1}{2ia^2g_0(a)} \left[U_{\mu\nu}(x) - U_{\mu\nu}(x)^\dagger - \frac{1}{N} \text{Tr}(U_{\mu\nu}(x) - U_{\mu\nu}(x)^\dagger) I \right],$$

and covariant finite differences give approximants to $D_\rho F_{\mu\nu}$. Gauge-invariant lattice composites $P_j^{(a)}$ are then mixed and renormalized by a finite matrix at each engineering dimension,

$$\mathcal{O}_P^{(a)} = \sum_{j: d_j \leq d_P} Z_{Pj}(a, \mu) P_j^{(a)}.$$

The required continuum statement is:

Bridge Lemma B4 (composite-field convergence). For every gauge-invariant differential polynomial $P(F, DF, \dots)$ there is a renormalized lattice approximant $\mathcal{O}_P^{(a)}$ such that all mixed Schwinger distributions containing finitely many such insertions converge in $\mathcal{S}'$ as $a \rightarrow 0$, and the limiting operators transform covariantly under the restored Euclidean group.

The stress tensor is obtained by differentiating an anisotropic/background-metric lattice action with respect to the metric and then renormalizing the finite set of operators allowed by symmetry. Its continuum Ward identity is

$$\partial^\mu T_{\mu\nu} = 0,$$

and after OS reconstruction

$$H = \int_{\mathbb{R}^3} T_{00}(0, \mathbf{x}) d^3x$$

in the standard quadratic-form sense on the energy domain.

At short distance, asymptotic freedom fixes the ultraviolet coefficients in products of local fields. The required operator-product statement is

$$\mathcal{O}_P(x)\mathcal{O}_Q(0) \sim \sum_R C_{PQ}^R(x, \mu, g(\mu))\mathcal{O}_R(0), \quad |x| \downarrow 0,$$

with $g(\mu) \rightarrow 0$ as $\mu \rightarrow \infty$ and Wilson coefficients having the perturbative asymptotic expansion determined by Yang–Mills renormalization.

Bridge Lemma B5 (short-distance matching). The renormalized continuum composite fields of B4 obey the Yang–Mills Ward identities and their short-distance Schwinger distributions have the asymptotic-freedom/OPE singular structure fixed by perturbative renormalization. In particular a conserved renormalized stress tensor exists.

B4 and B5 are the exact operator-theoretic form of the local-field and ultraviolet requirements in the Jaffe–Witten problem statement; they prevent a merely gapped lattice theory from being substituted for four-dimensional quantum Yang–Mills.

16. Reflection positivity and passage to the limit

At finite regulator the Wilson approximation is reflection positive; the heat-kernel formulation exhibits positivity directly through nonnegative character coefficients.

Let F be a polynomial in positive-time gauge-invariant smeared observables. Then

$$\langle \Theta F, F \rangle_{a,L} \geq 0.$$

If Bridge Lemma B2 gives convergence of every correlator entering this finite quadratic form, then

$$\langle \Theta F, F \rangle = \lim_{a \rightarrow 0} \lim_{L \rightarrow \infty} \langle \Theta F, F \rangle_{a,L} \geq 0.$$

Therefore reflection positivity is closed under the continuum limit once the Schwinger distributions converge.

The same argument passes permutation symmetry. Translation invariance passes from the translation-invariant regulators. Full Euclidean rotation invariance requires restoration from the hypercubic subgroup:

Bridge Lemma C1 (Euclidean restoration). The limiting Schwinger distributions are invariant under $E(4)$ rather than only the hypercubic subgroup, with all Lorentz-breaking irrelevant operators vanishing under the renormalized trajectory.

17. C: Osterwalder-Schrader reconstruction

Assume B1-B5 and C1 together with the standard OS regularity bounds. Then the limiting Schwinger family satisfies the reconstruction hypotheses. The OS quotient is formed from positive-time test polynomials by

$$(F, G)_{OS} = \langle \Theta F, G \rangle,$$

modulo null vectors and completed to a Hilbert space $\mathcal{H}$. Euclidean time translations generate a positive contraction semigroup

$$T(t) = e^{-tH}, \quad H \geq 0.$$

Spatial translations and rotations continue to unitary operators, analytic continuation produces the Poincaré representation, and the reconstructed local observable fields satisfy the corresponding Wightman-strength axioms.

Thus

$$\boxed{\{S_n\}_{n \geq 0} \xrightarrow{OS} (\mathcal{H}, \Omega, H, P_\mu, \mathcal{A}).}$$

18. D: spectral-gap transfer

For a local gauge-invariant observable A with $\langle \Omega, A\Omega \rangle = 0$,

$$C_A(t) = \langle \Omega, A e^{-tH} A \Omega \rangle = \int_{[0, \infty)} e^{-Et} d\rho_A(E),$$

where ρ_A is a positive spectral measure.

If the uniform regulator gap gives

$$|C_A^{(a, L)}(t)| \leq C_A e^{-m_* t}$$

with C_A stable under the limits, then

$$|C_A(t)| \leq C_A e^{-m_* t}.$$

Suppose $\rho_A([E_1, E_2]) > 0$ for some

$$0 < E_1 \leq E_2 < m_*.$$

Then positivity implies

$$C_A(t) \geq e^{-E_2 t} \rho_A([E_1, E_2]),$$

which cannot be bounded by $C_A e^{-m_* t}$ as $t \rightarrow \infty$. Hence

$$\rho_A((0, m_*)) = 0.$$

If local gauge-invariant vectors $A\Omega$ are dense in $\Omega^\perp$, then the bounded spectral projection $1_{(0,m_*)}(H)$ vanishes on a dense set and therefore vanishes identically. Thus

$$\text{Spec}(H) \subset \{0\} \cup [m_*, \infty).$$

19. Finite upper edge and nontrivial spectral support

The Jaffe–Witten definition also requires the supremal gap parameter to be finite. It is enough to construct one non-vacuum local state of finite energy. For a compactly supported gauge-invariant local operator A in the Hamiltonian form domain,

$$0 < \|(I - P_0)A\Omega\| < \infty,$$

and

$$\frac{\langle A\Omega, HA\Omega \rangle}{\|(I - P_0)A\Omega\|^2} < \infty$$

places spectrum at finite positive energy. The continuum nontriviality lemma B3 supplies the required non-vacuum local sector.

20. Arbitrary compact simple gauge group

The argument above is not special to $SU(3)$ at the structural level. For compact simple G :

1. Haar measure exists and is finite.
2. Peter-Weyl gives

$$L^2(G) = \widehat{\bigoplus_{\lambda \in \widehat{G}} V_\lambda \otimes V_\lambda^*}.$$

3. The Laplace–Casimir eigenvalue obeys $C_2(\lambda) > 0$ for every nontrivial irreducible representation.
4. Heat-kernel character coefficients are positive.
5. Wilson/character plaquette observables are local and gauge invariant.
6. The Schur theorem is representation independent.
7. The OS reconstruction and spectral-transfer arguments are group independent.

The group-dependent part is therefore concentrated in the uniform RG estimates of Bridge Lemmas A and B2. A complete “for any compact simple G ” proof supplies those estimates for each compact simple group, with no requirement that the numerical constant m_* be universal across different groups.

21. Positivity-bootstrap cross-check lane

Hermitian positivity, reflection positivity, and Schwinger–Dyson loop equations can be imposed simultaneously on finite sets of Wilson loops. The 2025 $SU(3)$ lattice bootstrap literature demonstrates rigorous expectation-value bounds from these constraints in 2D, 3D, and 4D. In this package that machinery is assigned as an independent finite-regulator cross-check lane:

Wilson loop algebra + reflection positivity + SD equations $\implies$ certified convex feasible region.

The natural next use is to bound the renormalized block parameters $(\eta_k, \delta_k, \Delta_k)$ entering Bridge Lemma A rather than merely bound the plaquette expectation.

22. Machine-verifiable results

The package runs the following independent checks:

- $SU(3)$ left/right covariance and fundamental Casimir;
- exact 18-dimensional finite Gauss-sector construction;
- positive interacting finite-workbench block gap;
- low-spectrum Feshbach-Schur isospectrality;
- exact spectral-parent inequality;
- positive self-adjoint transfer matrix and transfer-gap identity;
- positive-kernel preservation under coarse pullback;
- nonzero gauge-invariant finite spectral weight and exponential spectral bound;
- Wilson small-plaquette continuum scaling;
- one-loop asymptotically-free trajectory arithmetic;
- exact $SU(3)$ Peter-Weyl dimensions and Casimir values;
- fundamental tensor-product adjacency graph;
- heat-kernel positive coefficients and convolution semigroup coefficients;
- 4D hypercubic overlap/path-count constants;
- quadratic Schur contraction and infinite-step gap-floor certificate.

The current automated test suite returns **10/10 tests passed**. The prior independent finite-block verifier returns **11/11 checks passed**.

23. Jaffe-Witten proof ledger

The machine-readable file `results/jaffe_witten_obligation_ledger_v4.csv` maps every theorem to its dependencies. The shortest remaining $SU(3)$ proof chain is:

```

all-field UV RG control
↓
entry into the exact Schur gap basin
↓
 $\inf_{a,L} \Delta(a, L) = m_* > 0$ 
↓
thermodynamic + continuum Schwinger convergence
↓
Euclidean symmetry + OS axioms
↓
OS reconstruction
↓
 $\text{Spec}(H) \subset \{0\} \cup [m_*, \infty)$ 

```

The ledger deliberately separates a theorem already proved in the package, a standard theorem invoked from the literature, and a bridge lemma that must be discharged by the UCD/Wilson construction. That separation is part of the proof object: no finite-matrix eigenvalue is silently relabeled as a continuum mass.

24. Reproducibility

Install and run:

```
python -m pip install -r requirements.txt
python run_all.py
python -m pytest -q
```

The GitHub Actions workflow runs the same verification on each push and pull request. Every release file is hashed in MANIFEST.sha256.

25. Conclusion

The Jaffe–Witten problem has been converted into an explicit finite dependency graph. The compact-group proof regulator, gauge symmetry, Gauss sector, local Yang–Mills continuum asymptotics, Peter–Weyl finite verification hierarchy, positive Euclidean transfer structure, exact Feshbach–Schur identity, and a quantitative quadratic Schur gap-stability theorem are all written in one common normalization.

The new quantitative result is the exact summable Schur criterion

$$\Delta_\infty \geq \Delta_0 - \frac{2\delta_*}{A^2} \frac{(A\eta_0/\delta_*)^2}{1 - (A\eta_0/\delta_*)^2}.$$

This converts the infrared part of the mass-gap problem into measured/derived block constants. The continuum proof then rests on an equally explicit ultraviolet-to-block bridge and continuum Schwinger convergence estimates. Once those bridge lemmas are discharged, the OS and spectral-transfer sections give the required theory on $\mathbb{R}^4$ and

$$\text{Spec}(H) \cap (0, \Delta) = \emptyset, \quad \Delta > 0.$$

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